

Patrick Smith

Senior infrastructure engineer specializing in API integration, production troubleshooting, and customer-facing technical support



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Summary

For thirteen years I built and maintained production systems from the ground up. I architected APIs processing thousands of requests daily, deployed Linux infrastructure across AWS and Azure, optimized MySQL databases, and provided technical support to hundreds of users. I combined technical depth (debugging with strace/tcpdump/gdb, performance optimization, security incident response) with operational breadth (customer support, documentation, after-hours on-call).

I specialize in production-grade API integration and infrastructure reliability. Whether debugging a fail2ban outage using iptables, optimizing MySQL queries from $O(n^2)$ to linear time with spatial indexing, or reverse-engineering malicious PHP from process memory during a security incident, I solve problems by understanding systems end-to-end—from application code down to kernel behavior.

What drives me is building systems that work reliably under production pressure, debugging failures methodically when they don't, and helping people succeed whether they're customers stuck on a problem or teams collaborating on a solution.

Experience

Founding Engineer

May 2011 – Sept 2024

Home Remodel Buddy, San Francisco, CA

- Served as sole technical support engineer for 13 years managing ~740 contractors and advertising partners, providing end-to-end case management from triage through root cause analysis to resolution, including after-hours support (nights, weekends, holidays).
- Built Cerberus API gateway processing several thousand requests daily with 99.9% uptime, enabling contractors to submit leads via REST API with sub-250ms response times and detailed error reporting through custom HTTP headers.
- Integrated Stripe and PayPal payment processors with webhook event handling, idempotency keys for safe retries, and OAuth authentication flows, managing production payment processing for 13 years with zero payment data loss.
- Built Mercury customer portal to give 740 contractors self-service access to lead data and billing, eliminating support bottlenecks through real-time data access via PHP REST API and dual-mode authentication (sessions + API keys for integrations).
- Designed Hercules wholesale lead distribution system that algorithmically balanced historical return rates (2-25%), pricing, and load across partners, scaling revenue from \$25K to \$250K monthly per partner while maintaining SLA commitments.
- Integrated third-party REST/SOAP APIs (WhitePages, Spokeo, BeenVerified, Facebook, Twilio) with async batch processing, retry logic, and error handling to validate lead quality and automate customer communications.
- Optimized MySQL database queries from $O(n^2)$ haversine calculations to bounding box pre-computation using EXPLAIN analysis and spatial indexing, reducing geographic batch processing from hours to minutes across ~5M row database.
- Responded to WordPress security compromise by dumping malicious RAT from process memory via `/proc/{pid}/mem`, reverse-engineering obfuscated PHP (base64/eval chains), and completing full restoration in 4-5 hours with zero data loss.

Skills

- Programming
- Python

PHP

Bash

JavaScript

Lua
- Infrastructure & Cloud
- AWS

Docker

Git

Azure

Linux

CI/CD
- Networking
- VPC

DNS

VPN
- Databases
- MySQL

Memcached

Redis
- APIs & Monitoring
- REST

Grafana

SOAP

Other

Multi-Agent AI System Design

2025

Gridiron Dynasty (Open Source)

Architected collaborative AI agent system using Claude Code achieving 84% performance improvement (720s → 75s bootstrap) and 90% memory reduction (2.27GB → 224MB) through systematic optimization, parallel processing, and config caching. Designed specialized agent roles (Director, Architect, Engineer, Reviewer) with automated code review pipelines and comprehensive test coverage (83+ test cases).

Data Science

July 2025

HarvardX

- Debugged production incidents using Brendan Gregg's 60-second Linux performance methodology and advanced tools including strace for system call tracing, tcpdump for network inspection, lsof for file descriptor analysis, and gdb for process debugging.
- Debugged fail2ban production outage that blocked nginx-to-apache internal communication, identified issue using iptables -L, and restored service by whitelisting internal IPs to prevent recurrence.
- Deployed and managed Linux infrastructure across AWS (EC2, RDS, S3, CloudWatch, Route 53) and Azure, including VPCs, VPNs, and Docker containerization with custom blue-green deployment system using Git and atomic symlink switching for zero-downtime releases.
- Executed PHP 5.6 to 8.2 upgrade across four production systems in 2 weeks with zero downtime when hosting provider forced migration, resolving deprecated functions and library incompatibilities under pressure.
- Implemented monitoring using Grafana and Prometheus with custom PHP instrumentation, tracking server metrics (CPU, memory, I/O), API response times, error rates, and business metrics for proactive incident detection.
- Created comprehensive runbooks and technical documentation for API integrations, troubleshooting workflows, and system architecture—designed to be comprehensible six months later when context is forgotten.
- Collaborated with sales and product teams to prioritize feature development, translate business requirements into technical solutions, and communicate system capabilities to non-technical stakeholders.
- Mentored Founder's Dojo participants on API integration best practices, database optimization strategies, and production troubleshooting methodologies.

Education

A.S Business/Economics

2005-2007

Gavilan College, Gilroy, CA

Computer Science Coursework

2003-2005

California Polytechnic University, San Luis Obispo, San Luis Obispo, CA

CS109x: Introduction to Data Science with Python

July, 2025

HarvardX (Online Learning Initiative of Harvard University), <https://courses.edx.org/certificates/af05411c855d4ab7aaf9c8d90b29bd09>

Machine learning with sklearn, data analysis with Pandas/NumPy, regression and classification models, cross-validation, regularization, hypothesis testing, and statistical modeling

PH125.3x: Data Science: Probability

July, 2025

HarvardX (Online Learning Initiative of Harvard University), <https://courses.edx.org/certificates/73a1329507e642a0977384e4344c650e>

Probability theory, random variables, Monte Carlo simulations, Central Limit Theorem, expected values, standard errors, statistical inference foundations, and sampling models

Master Rank - Linux Troubleshooting

2025

SadServers.com, <https://sadservers.com/u/patrick/>

Completed advanced Linux troubleshooting scenarios including system debugging, performance analysis, networking issues, service failures, and production incident simulation

References

**Christopher
Levine, Director
of Eng**

Lyft

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**Barry
Willingham,
CEO**

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